American Computer Science League

2021 Finals ● Short Problems ● Junior Division

1. Boolean Algebra

How many ordered triples make the following expression TRUE?

$$A\,\overline{\overline{A}B \;+\; B\overline{C}}$$

A. 0
B. 3
C. 4
D. 6
E. None of the above

2. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{\overline{A}(A + \overline{B}) + B(\overline{A} + B)}$$

A. $\overline{A} + B$
B. $\overline{A}\,B$
C. $A\,\overline{B}$
D. $A + \overline{B}$
E. None of the above

3. Bit-String Flicking

Evaluate the following expression:

((LSHIFT-1 (NOT (RCIRC-2 01101) AND (LCIRC-2 01101)))
OR (RSHIFT-1 01101))

A. 10101
B. 10100
C. 01011
D. 01110
E. None of the above

4. Bit-String Flicking

Evaluate the following expression:

(NOT (01101 OR 01010) AND (01100 OR 10110))

A. 01010
B. 01111
C. 11110
D. 10000
E. None of the above

5. Recursive Functions

Find $f(25)$ given:

$$f(x) = \begin{cases} x + f(x-3) & \text{if } x \geq 15 \\ 2 \cdot f(x+2) & \text{if } 12 < x < 15 \\ x + 2 & \text{if } x \leq 12 \end{cases}$$

A. 140
B. 93
C. 58
D. 14
E. None of the above

6. Recursive Functions

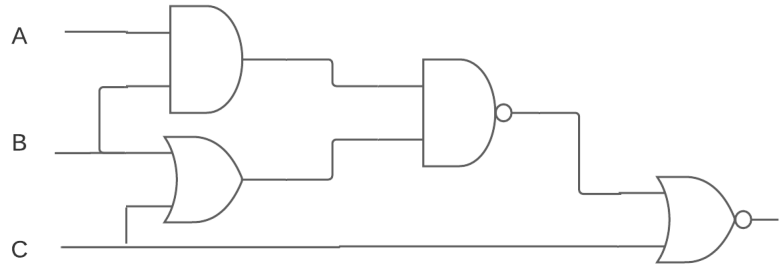
Given the following recursive function, what is the smallest value of y where $f(5, y) > 1000$?

$$f(x, y) = \begin{cases} 1 & if\ y = 0 \\ x \cdot f(x, y - 1) & if\ y > 0 \end{cases}$$

A. 4
B. 5
C. 10
D. 15
E. None of the above

7. Digital Electronics

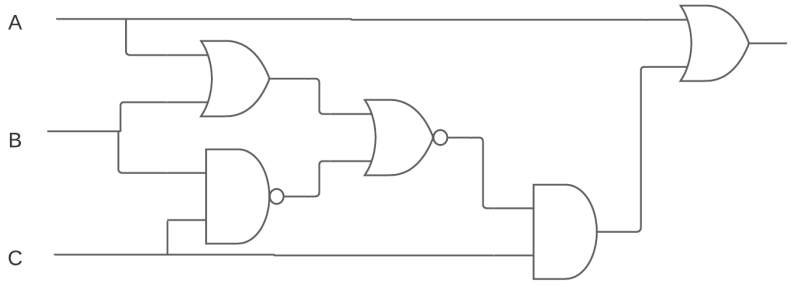
Simplify the Boolean expression represented by the circuit below:



A. $A\,B\,C$
B. $\overline{A}\,B\,C$
C. $A\,\overline{B}\,C$
D. $A\,B\,\overline{C}$
E. None of the above

8. Digital Electronics

How many ordered triples make the following circuit TRUE?



A. 0
B. 2
C. 4
D. 8
E. None of the above

9. Prefix-Infix-Postfix

Evaluate this prefix expression. (Note: all numbers are single digits.)

- + / + 1 8 3 * 4 3 ↑ 2 3

A. 6
B. 7
C. 8
D. 9
E. None of the above

10. Prefix-Infix-Postfix

Given the equation written in infix, convert it to postfix.

$$(x - h)^2 + (y - k)^2 = r^2$$

A. x h - 2 ↑ y k - 2 ↑ + r 2 ↑ =
B. x h - 2 ↑ y k - 2 ↑ r 2 ↑ + =
C. x h 2 ↑ - y k 2 ↑ - + r 2 ↑ =
D. x h - 2 ↑ + y k - 2 ↑ r 2 ↑ =
E. None of the above

11. Computer Number Systems

Given the year 2021, what is the next year that has 2 more 1s in its binary representation than 2021 has in its binary representation?

A. 2031
B. 2039
C. 2022
D. 2029
E. None of the above

12. Computer Number Systems

Evaluate the following and express the result in hexadecimal.

$AB_{16} + 74_8 - 1101_2$

A. EA_{16}
B. DA_{16}
C. $B3_{16}$
D. FB_{16}
E. None of the above

13. Data Structures

Build a binary search tree for:

NODINOSAURS

What is the depth of the tree?

A. 2
B. 3
C. 4
D. 5
E. None of the above

14. Data Structures

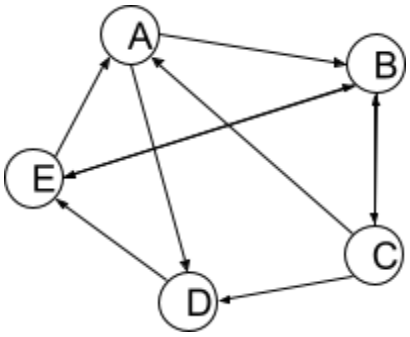
Given an initially empty stack and the following commands on the stack, what will the value of Z be?

PUSH(3), PUSH(7), PUSH(2), PUSH(4), X = POP(), Y = POP(), PUSH(X+Y), X = POP(), Y = POP(), PUSH(X-Y), X = POP(), Y = POP(), PUSH(X*Y), PUSH(9), PUSH(3), X = POP(), Y = POP(), PUSH(Y/X),
X = POP(), Y = POP(), PUSH(Y^X), Z = POP()

A. -9
B. 3
C. -27
D. 1
E. None of the above

15. Graph Theory

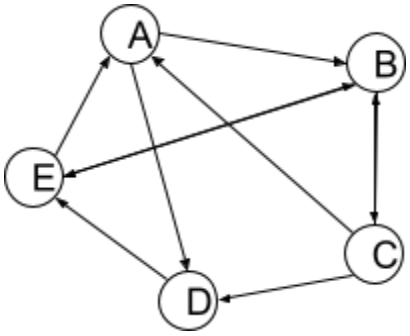
How many paths of length 2 are in the directed graph below?



A. 10
B. 15
C. 19
D. 21
E. None of the above

16. Graph Theory

How many cycles are there in the directed graph (same as #15)?



A. 7
B. 8
C. 9
D. 10
E. None of the above

17. What Does This Program Do?

Given a 6 x 8 two-dimensional array, a, that is populated by the numbers from 1 to 48 in row-major order, which of the following statement(s) inside the nested loops will print only those that are divisible by 3 or divisible by 5 exactly once?

```
for x = 0 to 5
        for y = 0 to 7
                …….
        next y
next x
```

A.
```
if a(x,y) % 3 == 0 then
    output(a(x,y))
if a(x,y) % 5 == 0 then
    output(a(x,y))
```

B.
```
if a(x,y) % 15 == 0 then
    output(a(x,y))
```

C.
```
if a(x,y) % 3 == 0 or
a(x,y) % 5 == 0 then
    output(a(x,y))
```

D.
```
if a(x,y) % 3 == 0 and
a(x,y) % 5 == 0 then
    output(a(x,y))
```

E. None of the above

18. What Does This Program Do?

What is the output for the following program?

```
r = 1
p = 1
t = 1
for x = 1 to 8
     r = r * x
next x
for x = 1 to 3
     p = p * x
next x
for x = 1 to 5
     t = t * x
next x
output r / (p * t)
```

A. 1
B. 42
C. 48
D. 56
E. None of the above

19. What Does This Program Do?

Given that array **arr** contains the values below, how many values will be output by the following program?

11	16	24	39	43	56	69	81	97

```
for x = 0 to 8
     check = 1
     for y = 3 to int(arr(x)/2) step 2
          if arr(x) % y == 0 then
               check = 0
          end if
     next y
     if check == 1 then
          output(arr(x))
     end if
next x
```

A. 2
B. 3
C. 4
D. 5
E. None of the above

20. What Does This Program Do?

Using ACSL code, which of the following statements can be used where the blank line is to produce the following output in the given program?

Output:

```
F
_FI
L_FIN
SL_FINA
CSL_FINAL
ACSL_FINALS
```

```
s = "ACSL_FINALS"
for x = (len(s)-1)/2 to 0 step -1
     output(____________)
next x
```

A. s[int(len(s) / 2) - x : x]
B. s[x+1 : len(s) - x]
C. s[x : len(s) - x + 1]
D. s[x : len(s) - x - 1]
E. None of the above